

Factors Influencing the Condition of Sewer Pipes: State-of-the-Art Review

Mohammadreza Malek Mohammadi, Ph.D., M.ASCE¹; Mohammad Najafi, Ph.D., P.E., F.ASCE²;
Sharareh Kermanshachi, Ph.D., P.E., M.ASCE³; Vinayak Kaushal, Ph.D., A.M.ASCE⁴;
and Ramtin Serajiantehrani, Ph.D., A.M.ASCE⁵

Abstract: Wastewater infrastructure systems deteriorate over time due to a combination of physical and chemical factors. Failure of these critical structures can cause major social, environmental, and economic impacts. To avoid such problems, several researchers attempted to develop infrastructure condition assessment methodologies to maintain sewer pipe networks at desired condition. Sewer condition prediction models are developed to provide a framework to forecast future conditions of pipes and to schedule inspection frequencies. Yet, utility managers and other authorities are often challenged with identifying the optimal timeline for inspection of sewer pipelines. **Frequent inspection of sewer networks is not cost-effective due to limited time, expensive assessment technologies, and large inventories of pipes.** Therefore, the objective of this state-of-the-art review is to study progress over the years in developing condition prediction models and investigating the potential factors affecting the condition of sewer pipes. Published papers for prediction models from 2001 through 2019 were identified and analyzed. Also, this study conducts a comparative analysis of the most common condition prediction models such as artificial intelligence (AI) and statistical models. The literature review suggests that, out of 20 independent variables studied, pipe age, diameter, and length are the most significant contributors to the deterioration of sewer systems. In addition, it can be concluded that AI models reduce uncertainty in current condition prediction models. Furthermore, the most appropriate prediction models for development are those that are capable of accurately finding nonlinear and complex relationships among variables. This study recommends the use of more environmental and operational factors—e.g., soil type, bedding material, flow rate, and soil corrosivity—and advanced data mining techniques to develop comprehensive and accurate condition prediction models. The findings of this study are intended to guide practitioners in developing customized condition assessment models for their agencies that can save millions of dollars through optimized inspection timelines and fewer incidents. **DOI: 10.1061/(ASCE)PS.1949-1204.0000483.** © 2020 American Society of Civil Engineers.

Author keywords: Sewer condition prediction; Sewer pipe prioritization; Deterioration model; Pipe condition assessment; Asset management.

Introduction

Underground infrastructure systems span thousands of miles and form a significant part of the total infrastructure of the United States (Najafi and Gokhale 2005). Sanitary sewers, as a part of wastewater

infrastructure systems, are designed to collect sewage from domestic, industrial, and commercial users and convey it to treatment plants. Most sewer systems are gravity sewers, which transfer the flow based on a slope. There are over 800,000 mi of public sewer pipes and 500,000 mi of private sewer laterals in the United States. Approximately 240 million Americans are connected to 14,748 treatment plants for wastewater treatment. By 2032, it is estimated that 56 million more people will use centralized treatment plants (ASCE 2017).

Most of the US wastewater infrastructure is over 100 years old, and the combination of aging, chemical, and environmental factors cause at least 23,000–75,000 sanitary sewer overflows per year (EPA 2004). **The latest infrastructure report card published by the American Society of Civil Engineers (ASCE 2017) gives the wastewater infrastructure a “D+” grade, which indicates poor condition.** ASCE indicated that water and wastewater systems in the United States are clearly aging and, to keep up with the need, a capital funding gap of \$150 billion will be needed by 2025 (ASCE 2017). Furthermore, the US population is increasing and shifting geographically. This demographic shift will require investment in new infrastructure and maintenance of existing infrastructure in areas of decreasing population with limited budgets (EPA 2007).

Ana et al. (2009) suggested that the best way to forecast pipe failure and deterioration time is to develop probability-based condition prediction models based on actual inspection databases. In addition, it is essential to identify the influence factors associated with sewer pipe deterioration to develop a reliable condition prediction model, which is a key component of sewer asset

¹Doctoral Graduate, Center for Underground Infrastructure Research and Education, Dept. of Civil Engineering, Univ. of Texas at Arlington, P.O. Box 19308, Arlington, TX 76019 (corresponding author). Email: mohammadreza.malekmohammadi@mavs.uta.edu

²Professor and Director, Center for Underground Infrastructure Research and Education, Dept. of Civil Engineering, Univ. of Texas at Arlington, P.O. Box 19308, Arlington, TX 76019. Email: najafi@uta.edu

³Assistant Professor, Dept. of Civil Engineering, Univ. of Texas at Arlington, 438 Nedderman Hall, 416 Yates St., Arlington, TX 76019. Email: sharareh.kermanshachi@uta.edu

⁴Postdoctoral Research Associate, Center for Underground Infrastructure Research and Education, Dept. of Civil Engineering, Univ. of Texas at Arlington, P.O. Box 19308, Arlington, TX 76019. ORCID: <https://orcid.org/0000-0001-7922-2746>. Email: vinayak.kaushal@mavs.uta.edu

⁵Doctoral Graduate, Center for Underground Infrastructure Research and Education, Dept. of Civil Engineering, Univ. of Texas at Arlington, P.O. Box 19308, Arlington, TX 76019. ORCID: <https://orcid.org/0000-0001-9340-1142>. Email: ramtin.serajiantehrani@mavs.uta.edu

Note. This manuscript was published online on June 27, 2020. Discussion period open until November 27, 2020; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Pipeline Systems Engineering and Practice*, © ASCE, ISSN 1949-1190.

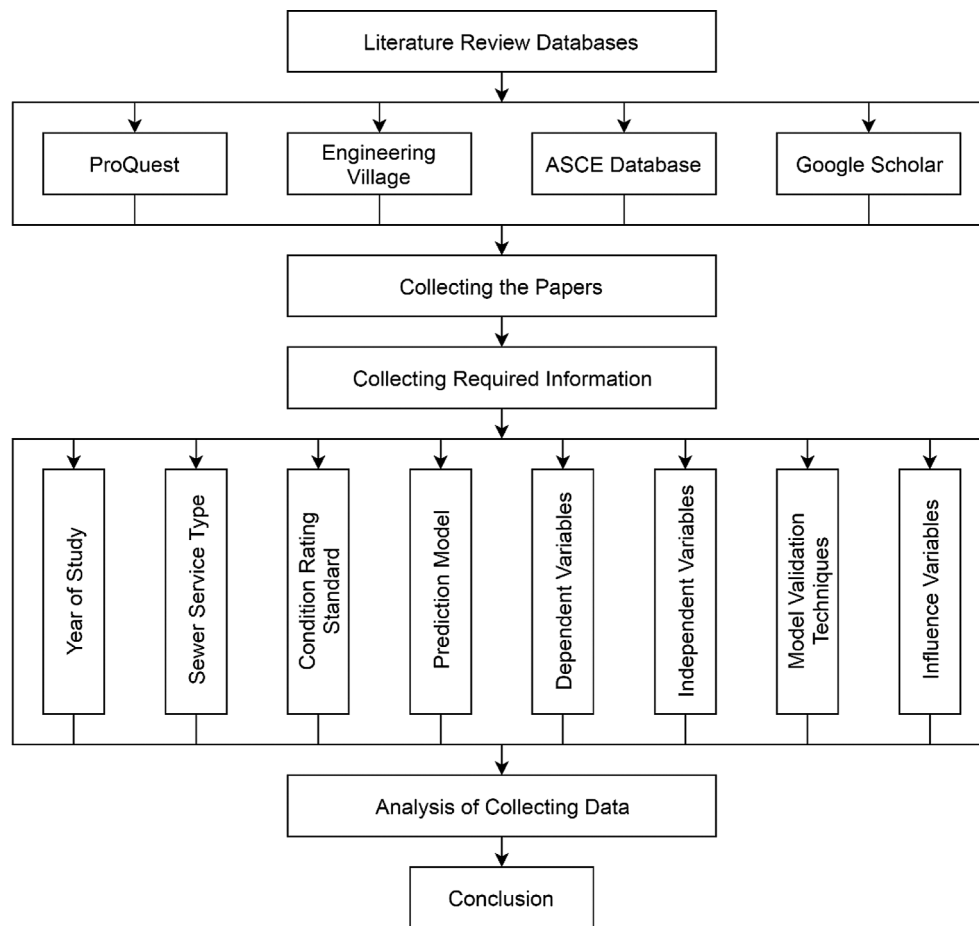


Fig. 1. Research methodology.

management systems (Ana et al. 2009). Identifying the significant factors will reduce number of required variables in pipe inventories and data collection costs, and will increase the accuracy of a condition prediction model (Kley and Caradot 2013).

By summarizing published literature, the objective of this paper is to investigate prediction models that forecast conditions of sewer pipes with consideration of the physical, environmental, and operational factors in the deterioration process. In addition, the effect of influence variables affecting deterioration of sewer pipes will be evaluated. Areas of contention and uncertainties will be discussed, and recommendations for future research are provided.

Methodology

To fulfill the objectives of this study, several journal articles, conference papers, dissertations, and research reports were studied. As demonstrated in Fig. 1, published papers were collected from various databases, such as ProQuest, Engineering Village, ASCE Database, and Google Scholar, that analyzed prediction models from 2001 through 2019. All of these papers were reviewed, and essential data—year of study, sewer service type, condition rating standards used, prediction models, dependent and independent variables, model validation techniques, and variables influencing deterioration of sewer pipes—were collected. A number of analyses regarding the effect of influence variables on deterioration of sewer pipes were performed for each category to gain better understanding about their influence on development of condition prediction models.

Sewer Condition Prediction Models

Condition prediction models can be used to forecast condition rating of sewer pipes by using information obtained from inspection databases. Prediction models can perform an essential role in generating a prioritized inspection plan, and provide valuable information to forecast short- and long-term behavior of sewer pipes. In general, utility companies and municipalities can forecast the future condition of their assets by generating deterioration models to identify the pipes that require maintenance, rehabilitation, and replacement. The primary objective of sewer condition prediction models is to apply an appropriate mathematical technique to estimate future condition states of sewer pipes. Additionally, condition prediction models can identify significant factors affecting deterioration of the pipes (Salman and Salem 2012).

The current condition of sewer pipes is often assessed through inspection techniques; however, understanding the future condition of pipe systems requires a comprehensive deterioration model. Most of the sewer prediction models are developed through datasets obtained from closed-circuit television (CCTV) inspection to forecast the failure time of pipes based on condition rating standards (Malek Mohammadi 2019).

Deterioration models for sewer pipelines are classified into different categories. Deterministic, probabilistic, statistic, physical, and AI models are the most common techniques that have been used in previous studies (Malek Mohammadi et al. 2019a; Altarabsheh 2016; Kley and Caradot 2013; Tran 2007; Yang 2004). Kley and Caradot (2013) divided the deterioration models into

deterministic, probabilistic, and soft computing methods. Yang (2004) used physical, AI-based, and statistical categories for condition prediction. Tran (2007) suggested deterministic and statistical models as a model-driven type and AI-based models as a data-driven type. Typically, model-driven data are defined by the expert, whereas the sample data demonstrate the structure of models in the data-driven type.

Additionally, Altarabsheh (2016) classified the deterioration models into deterministic and probabilistic models. It is obvious that accuracy of model prediction is highly dependent on selecting the proper modeling techniques for sewer deterioration. A deterministic model is the inherent lack of randomness or stochasticity, and is widely used when relationships between components are certain (Tran 2007). This model assumes that the output can be exactly predicted by given input variables. For example, if the current state of the pipeline system is perfectly known, it is possible to predict the exact future condition of network by generating a deterministic model.

The basic explanation of a statistical model is a random variable X , which represents a quantity whose outcome is uncertain. In statistical models, the probabilistic nature of historical data is used to describe the model output as a random variable. In any statistical analysis, estimates are best guesses based on the condition of given historical data (Coles 2011).

Charniak and McDermott (1985) define AI as “the study of mental faculties through the use of computational models.” In AI models, the dependent variables are classified from a set of independent variables by learning from the available data. These models are appropriate to estimate ordinal condition ratings or

nonlinear deterioration behaviors (Russell and Norvig 2010). The use of AI and statistical models to predict sewer pipe deterioration has increased; physical models are less commonly used due to their complexity. Furthermore, deterministic, probabilistic, and statistical models can be in the same group based on their statistical nature.

Thus, existing sewer deterioration models can be classified into two groups: statistical and AI models, as shown in Fig. 2. **Several statistical models such as logistic regression, Markov chain, ordinal regression, and linear regression have been used in previous studies to forecast the condition of sewer pipes. With the growth in AI models, machine learning and neural network models have become popular because they can detect more complex relationships between variables.**

Table 1 presents sewer condition prediction models using different statistical and AI models. This table indicates the number of data points and condition rating standards used in previous studies to develop condition prediction models for sewer pipes. These models are expected to provide decision-making criteria for utilities and municipalities regarding the development of inspection timelines, rehabilitation strategies, and section-level risk assessment models.

Condition Rating Methods for Sewer Pipes

The condition of sewer pipes is generally specified through a process of condition assessment forming. The historical background of the sewer condition assessment protocols traces back to 1977, when the Water Research Centre (WRC), in the United Kingdom, started

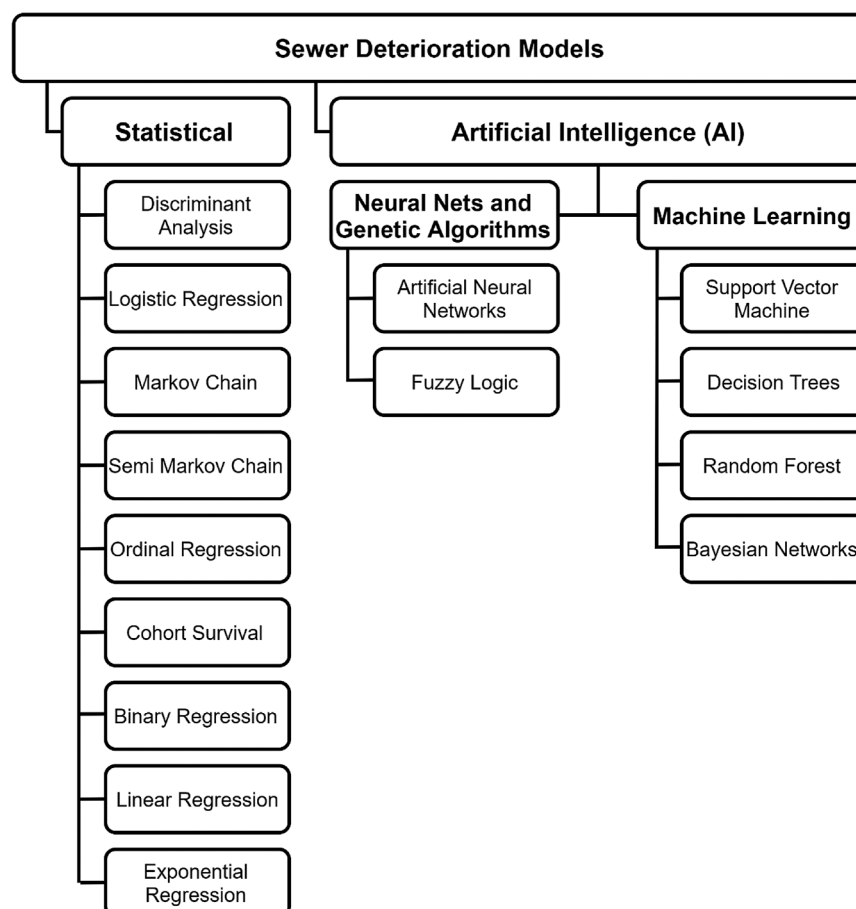


Fig. 2. Classification of sewer deterioration models.

Table 1. Sewer condition prediction models

Author and year	Model	Number of data	Condition rating standard
Davies et al. (2001)	Logistic regression	12,000	WRc
Ariaratnam et al. (2001)	Logistic regression	748	WRc
Wirahadikusumah et al. (2001)	Markov chain	—	Other
Micevski et al. (2002)	Markov chain	497	SEWRAT
Najafi and Kulandaivel (2005)	Neural network	—	PACP
Tran et al. (2006)	Neural network	583	WSAA
Koo and Ariaratnam (2006)	Logistic regression	579	PACP
Tran et al. (2007)	Neural network, multiple discrimination analysis	150	WSAA
Chughtai and Zayed (2008)	Linear regression	—	WRc
Gat (2008)	Markov chain	5,262	DWA
Ana et al. (2009)	Logistic regression	1,316	NEN3399
Tran et al. (2009)	Neural network, ordered probit model	417	WSAA
Khan et al. (2010)	Neural network	200	WRc
Mashford et al. (2011)	Support vector machine	1,441	Other
Lubini and Fuamba (2011)	Logistic regression	459	PACP
Salman and Salem (2012)	Ordinal regression, logistic regression, binary regression	11,373	PACP
Sousa et al. (2014)	Neural network, support vector machine, logistic regression	745	PACP
Harvey and McBean (2014)	Random forest, decision tree, support vector machine	1,825	WRc
Bakry et al. (2016a)	Multiple regression	84	PACP
Gedam et al. (2016)	Linear regression	155	PACP
Kabir et al. (2018)	Bayesian logistic regression	12,728	PACP
Laakso et al. (2018)	Random forest, binary logistic regression	6,700	EN-13508-2
Malek Mohammadi et al. (2019b)	Binary logistic regression	21,000	PACP

a 5-year research project to implement a methodology to assess the condition of sewer pipes based on a general coding system. The world's first *Sewerage Rehabilitation Manual (SRM)* was published in 1980 by WRc, which later became the main reference for further development of sewer condition assessment protocols (Chughtai and Zayed 2011; Rahman and Vanier 2004).

The condition rating is used to objectively evaluate the current condition of sewer pipes. This rating standard is based on the results of pipe inspection normally carried out using CCTV (Ana et al. 2009). Structural condition and operational condition are two common types of pipe condition categories (Chughtai and Zayed 2008). The structural condition evaluates the pipe defects, the physical strength of a pipe, and the capability of the pipe to resist external loads. The operational condition indicates the ability of the pipe to meet its service requirements. The result of structural conditions can be used to determine the necessity of pipe rehabilitation or replacement, whereas the operational condition of a pipe indicates the need for cleaning and maintenance (Opila 2011).

Numerous methodologies have been developed to score the condition of buried sewer pipes in different countries, such as WRc in United Kingdom, Pipeline Assessment Certification Program (PACP) in the United States, and those of the National Research Council (NRC) in Canada and Water Services Association of Australia (WSAA) (Moteleb 2010). In general, condition prediction scales are classified by discrete or finite scale values of relatively limited ranges (Baur and Herz 2002). For example, WSAA and PACP assess the condition of pipes on a scale of 1 (new) to 5 (failed condition). As presented in Table 1, WRc and PACP are the most common standards used to develop condition prediction models for sewer pipes based on the result of the reviewed literature in this paper. Fig. 3 illustrates frequency of condition rating standards used in previous studies from 2001 to 2019.

Factors Influencing Deterioration of Sewer Pipes

Deterioration of sewer pipes is a very complex process that is affected by several factors rather than just a single factor

(Lindner 2008; Opila 2011; Atique 2016). Selecting proper construction and equipment can affect performance of underground buried infrastructures (Jalalediny Korky et al. 2019). Davies et al. (2001) categorized the factors that influence structural deterioration of rigid pipes into three groups: construction, operational, and environmental factors. Typically, agencies and municipalities have enough physical data, but information about environmental and operational factors is often unavailable (Salman 2010). According to Kley and Caradot (2013), it is important to identify factors that influence the deterioration of sewer pipes for following reasons:

- Data collection is a very expensive process during condition assessment, and gathering all the pipe information is not a cost-effective approach. Identification of the significant factors

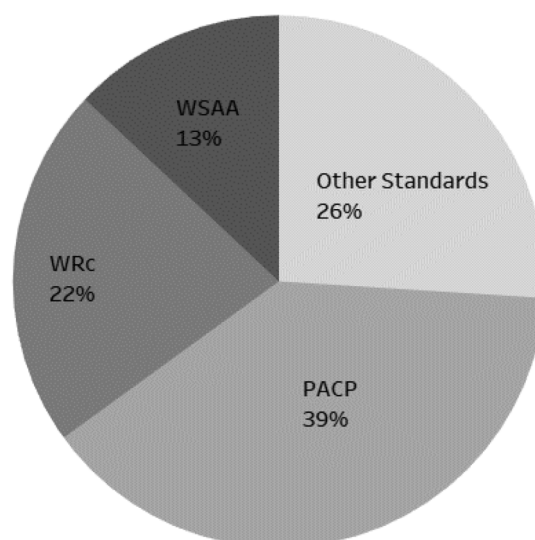
**Fig. 3.** Frequency of condition rating standards in previous studies.

Table 2. Factors affecting sewer pipe deterioration

Physical factors	Environmental factors	Operational factors
End invert elevation	Backfill type	Blockages
Installation method	Bedding material	Burst history
Joint type	Ground movement	Debris
Pipe length	Groundwater level	Flow velocity
Pipe shape	pH	Hydraulic condition
Pipe slope	Road type	Infiltration/exfiltration
Pipe age	Root interference	Previous maintenance
Pipe depth	Soil corrosivity	Sediment level
Pipe material	Soil fracture potential	Sewer type
Pipe size	Soil moisture	—
Start invert elevation	Soil type	—
	Soil sulfate level	—
	Traffic characteristics	—
	Vehicle flow	—

decreases the number of required data in pipe inventories and reduces data collection costs.

- High prediction accuracy can be achieved when significant factors are used in the model.

In general, factors influencing the deterioration of sewer pipes can be classified into physical, environmental, and operational. Table 2 the presents the most frequent factors or variables used in previous studies to predict the condition of sewer pipes. The effects of the most frequent factors used in previous studies will be presented in following sections.

Pipe Age

Pipe age is normally the difference between pipe installation year and date of inspection. The aging of pipe begins the minute the sewer pipe is installed, and aging is one of the most important factors in pipe deterioration (Kulandaivel 2004). The bathtub curve is a plot that determines the rate of pipe failure depending on the age of the pipe (Singh and Adachi 2013). The bathtub curve involves three distinct phases, as shown in Fig. 4.

The first phase is the early life period with high failure rates, exhibiting failures right after installation. In this phase, the failures can occur because of human error, pipe damage during construction and installation, and inappropriate pipe material. The second phase shows the useful life of the pipe, and the frequency of failure rate is

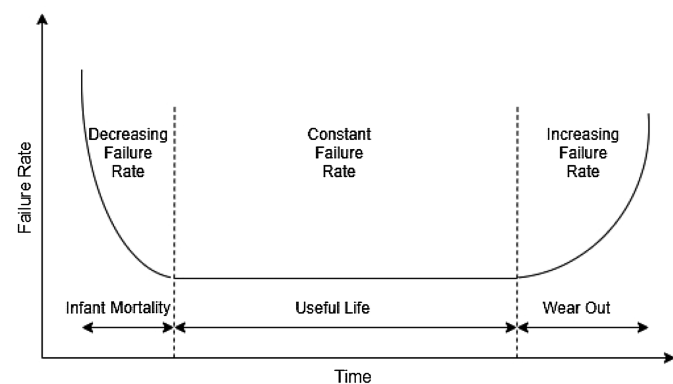


Fig. 4. Theoretical bathtub curve of buried pipe. (Data from Singh and Adachi 2013.)

very low and almost constant in this phase. Several random phenomena such as extreme heavy loading, earth movement, settlement, or third-party interference can be the result of failures in the second phase. And, in the third phase (wear-out life) the frequency of failure is high due to the pipe deterioration and the aging process (Singh and Adachi 2013).

Most of the condition prediction models developed in previous studies show that pipe age has a significant relationship with deterioration of sewer pipes. Ariaratnam et al. (2001) stated that pipe age significantly affects deterioration of sewer pipes due to the consequence of the pipe aging process. Jeong et al. (2005) and Ana et al. (2009) had similar findings, and indicated that the deterioration rate is lower during the early years of pipe service life and higher during the later years. Khan et al. (2010) specified that deterioration of sewer pipes does not start right after the installation process but arises after a certain period of time.

Lubini and Fuamba (2011) demonstrated that with aging of sewer pipes, root intrusion increases and pipe roughness increases gradually. Since there is a direct relationship between pipe roughness and friction factor, the hydraulic performance of the pipe will decline and the likelihood of pipe deterioration will increase. Pipe age was found significant in prediction models developed by Salman and Salem (2012), Kabir et al. (2018), and Laakso et al. (2018).

In contrast, Davies et al. (2001) found that pipe age is not a significant variable in the sewer pipe deterioration model; however, their dataset did not include the age of each individual sewer pipe, and property age was the reference value of pipe age. Tran et al. (2006) presented a similar result, stating that pipe age is not a significant factor because the structural deterioration of sewer pipes depends on the combined effects of various factors.

Pipe Material

Sewer pipes constructed with different materials have different reactions to environmental factors, such as soil type, water table, and others. (Salman 2010). For example, concrete pipes are highly resistant to abrasion, and clay pipes act very well against acids. Plastic pipes, such as polyvinyl chloride (PVC) or high-density polyethylene (HDPE), resist acidic and alkaline wastes; however, they can suffer excessive deformations under loading (Singh and Adachi 2013). Pipe material can be used as an independent variable during development of condition prediction models, and the results of the model can reveal whether this variable is significant.

Davies et al. (2001) found that pipe material is a significant variable and that there is a direct relationship between the deterioration of sewer pipes and pipe material. Micevski et al. (2002) noted that concrete pipes are stronger and more durable than clay pipes based on the results of their Markov model. Ana et al. (2009) indicated that concrete pipes exhibited better behavior in the model than brick and clay pipes. One probable reason for difference in aging behaviors of pipes is due to the production procedures of the pipes. Concrete pipes are typically constructed offsite in a controlled environment, resulting in high quality and integrity; however, brick pipes are usually constructed in situ and in different environmental conditions, and poor workmanship can affect quality.

Pipe material also had a significant effect in the model developed by Lubini and Fuamba (2011). They found that reinforced concrete pipes are the most resistant to deterioration compared to other pipes because of the reinforcing steel that makes the conduit strong enough to prevent structural deterioration. Bakry et al. (2016a) demonstrated that vitrified clay pipes behave better than asbestos cement and reinforced concrete pipes in their model. In the prediction model developed by Laakso et al. (2018), concrete

and high-density polyethylene pipes were found significant. A possible explanation for the different behavior of pipe material in their study was that the quality of certain batches of HDPE pipes was deficient.

In contrast, Jeong et al. (2005) stated that pipe material was not a significant variable in their study (which they noted could be due to their use of fewer data to develop the prediction model), nor was class imbalance. In general, a better pipe behavior can be predicted if separate models are generated for different pipe materials.

Pipe Diameter

Numerous studies investigated the relationship between sewer size and deterioration of the pipes, and the results are contradictory. Some condition prediction models identified that sewer deterioration rate decreases with increasing the diameter, whereas other studies found that smaller diameter pipes have better behavior and less failure. Pipe diameter was found to be a significant factor affecting deterioration of sewer pipes in several prediction models. Ariaratnam et al. (2001) indicated that, in their model, the likelihood of a pipe being in deficient condition is higher when the diameter increases. Davies et al. (2001) found that the risk of rigid sewer pipes being in poor condition decreases significantly with increasing diameter, and larger sewers are at a lower risk than small ones; the reason, they explained, could be that the structural design of rigid sewer pipes is restricted to the cross section of the pipes and the ring or crushing stress experienced.

Micevski et al. (2002) found that deterioration of smaller pipes was greater than the larger pipes. A probable explanation is that the pipe designers underestimate the required depth of cover and loading traffics for the smaller pipes. Tran et al. (2009) indicated that larger pipes are often buried deeper and more appropriate design and construction crews are used to install them; therefore, larger-diameter pipes are more resistant to deterioration based on the result of the developed prediction model.

Lubini and Fuamba (2011) determined that with the occurrence of obstacles in the conduit, segments with larger diameter are still able to convey wastewater and small diameters are more likely to deteriorate due to loss of hydraulic flow. Salman and Salem (2012) and Bakry et al. (2016b) obtained the same result and, in their prediction model, larger pipes behave better than smaller pipes and have lower deterioration rates.

In contrast, Jeong et al. (2005) stated that larger pipes are more likely to deteriorate because they have more surface area exposed to sewage and surrounding soil areas. The larger pipes are more at risk of damage because they are heavy and bulky, and it is difficult to install them accurately. Laakso et al. (2018) determined that pipes of 30 and 150 cm (12- and 60-in.) diameter were in better condition due to more careful supervision during design and installation phases.

Tran et al. (2006) and Ana et al. (2009) found that pipe diameter is not a significant variable in their models; however, they demonstrated that larger pipes are usually buried deeply and have lower deterioration rate than smaller ones.

Pipe Length

In practically all sewer pipe inventories, length of pipes is recorded as pipe segment that is manhole-to-manhole length of multiple pipe sections. Typically, longer manhole-to-manhole sewer pipe segments have higher deterioration rates because the probability of defects is greater in longer pipes. However, in previous studies, a dual behavior was found in the effect of pipe length to deterioration rate.

Davies et al. (2001) indicated that the risk of pipe being in poor condition declines when individual pipe sections were longer [more

than 1.5 m (5 ft)]. Pipe joints are the main source of infiltration in pipelines, which results from the movement of soil into a sewer and lack of support, leading to structural instability. Longer individual pipe sections means that the number of joints per manhole-to-manhole length of sewer pipes (pipe segment) is reduced, and thus the risk of infiltration and joint defects is also reduced. Jeong et al. (2005) indicated that longer sewer pipe segments are less likely to deteriorate than shorter ones. They used a dataset including large-size pipes, and a probable explanation could be that larger pipes are less likely to have blockages, and therefore the sewage can be conveyed more easily.

Ana et al. (2009) explained that the risk of pipe deterioration increases when sewer pipe manhole-to-manhole segments are longer, possibly because longer pipes have more joints, and joints are especially vulnerable to failure. Joint defects are a common in sewer systems, and they increase the probability of failure. Additionally, longer pipes are more vulnerable to blockages and sediment deposition, which facilitate the deterioration of sewer pipes. Khan et al. (2010) found a dual behavior in the condition of pipes with respect to changes in pipe length—that is, pipe segments smaller than 70 m (230 ft) have no effect on the condition of sewer pipes, while pipe segments longer than 70 m (230 ft) increase the rate of deterioration due to the density of end joints which are a source of break, infiltration, and exfiltration.

Salman and Salem (2012) determined that longer pipes behave better in a sewer network because, as the length of the pipe increases, the level of exposure to deteriorating factors also increases. Laakso et al. (2018) found that sewer pipes longer than 40 m (131 ft) deteriorate faster than other pipes in a network, which can be explained by the higher potential of defects and bending stress in longer pipes. Additionally, lateral connections are potential causes of structural damage, and longer pipes have more lateral connections.

Pipe Depth

Several factors such as soil type, water table, pipe material, pipe diameter, and regulations must be considered to identify the appropriate depth for sewer pipes. The results of investigating the effect of depth on the deterioration of sewer pipes are contradictory in different prediction models. Khan et al. (2010) indicated that pipe depth is a significant variable in their prediction model, and that an increase in depth has a negative effect on the sewer pipe condition level. The rational reason for this behavior could be that deeper pipes have greater dead load over them and a higher probability of encountering the groundwater table. In contrast, pipe depth was not a significant variable in the prediction model developed by Davies et al. (2001).

This is not to say that sewer depth does not affect the deterioration of pipes when considered on its own, but in data analysis based on the features of pipe datasets, there is not necessarily a direct relationship between pipe depth and condition level of sewer pipes. Tran et al. (2006) and Ana et al. (2009) also reported that pipe depth was not a significant factor in their prediction models. Generally, shallowly buried pipes would be subjected to more defects and higher deterioration rate due to surface load, illegal connections, and tree root intrusion. Additionally, more cover depth above the pipes decreases the effect of surface factors such as road traffic and road maintenance/construction activities.

Salman and Salem (2012) had the same findings. Among the eight independent variables used in their model, pipe depth was the only insignificant variable in the model. Laakso et al. (2018) demonstrated that installation depth between 1.8 and 3 m (6 and 10 ft) was correlated with poor condition level in their study, and they recommended minimum depth of 1.5 m (5 ft) due to winter frost conditions.

Pipe Slope

Sewer pipes with flat slopes have lower velocities, and thus wastewater stays longer inside the pipe. The longer the wastewater remains in the sewer pipes, the more likely hydrogen sulfide gas will be generated inside the sewer. Hydrogen sulfide can be converted to sulfuric acid, which attacks cementitious pipes, such as concrete and mortar, and increases the rate of corrosion inside the sewer pipes (Ana et al. 2009; Ayoub et al. 2004). Similarly, Baur and Herz (2002) indicated that sediment deposition and clogging occur more in pipes with flatter slopes, and risk of deterioration is higher in these pipes.

Jeong et al. (2005) indicated that when the pipe slope is steeper, the probability of deterioration is higher in sewer segments. Faster flow rate and lower stability are the probable causes of the higher deterioration rate, based on the results of this study. Tran et al. (2006) had similar findings, and suggested that pipes with steeper slopes are more vulnerable to defects due to voids in the soil, soil movement, and pipe joint defects. The prediction model developed by Salman and Salem (2012) revealed the significance of pipe slope and, according to their results, pipes at steeper slopes are more likely to deteriorate due to stability issues and high flow rate.

Laakso et al. (2018) determined that negative and very low slopes were the most harmful conditions for sewer pipes based on the result of their prediction model. Negative slopes and extremely low slopes cause inadequate rinsing, which can lead to debris accumulation and blockages. In summary, pipe slope was found an insignificant variable based on the results of Tran et al. (2006), Ana et al. (2009), Sousa et al. (2014), and Kabir et al. (2018).

Sewer Type

Sewer pipes can be classified into separate sanitary, storm sewer systems, and combined sewer systems. In a combined sewer system, a single pipe is used to transport domestic, commercial and industrial wastewater, and storm water to a selected disposal location. The concept of separate sanitary sewer and storm sewer system is to manage storm water and sanitary wastewater separately. In this method, two separate pipes are used to convey domestic, commercial and industrial wastewater, and storm water to a selected disposal location.

O'Reilly et al. (1989) found that the rate of deterioration is higher in combined sewer systems than in sanitary sewers. They argued that, generally, combined sewer systems have shallower installation than separate systems, and the flow fluctuation is higher in combined systems. Davies et al. (2001) and Baur and Herz (2002) showed a lower deterioration rate in combined sewers because these system types underwent more planning and engineering effort during their construction. Ariaratnam et al. (2001) demonstrated that waste type is a significant variable, and that sanitary sewers are found to have the greatest effect on pipe deficiency, followed by storm and then combined sewer systems.

Salman and Salem (2012) claimed that sanitary sewer pipes are more resistant to deterioration than combined sewers. This outcome can be explained by the higher potential for soil loss, infiltration, and exfiltration in combined sewers due to the high flow velocity during rainfall events.

Loading on Sewer Pipes

Obviously, a sewer pipe can be affected by applied load from the surface. Land use and traffic on the surface above the sewer pipe affect the magnitude of surface loading carried to the pipe. It is very difficult to measure or estimate the magnitude or frequency of surface loads because they vary in time (Kley and Caradot 2013).

Sewer pipes can be subjected to large one-time loads, such as surface construction, ground utility construction, landslides and earthquakes, or to small cyclic loads with hourly, daily, or seasonal frequency, such as bus stops, traffic, and maintenance activities (Ashoori et al. 2017).

It should be mentioned that the influence of road type on the deterioration of sewer pipes was investigated in a few studies. Davies et al. (2001) determined that sewer located under rural main roads and sidewalks were at a significantly lower risk of being in poor condition than those pipes located under urban main roads. The main reason for this difference may be that pipes are more affected by a heavier traffic load in urban areas. Pipe location was also found significant based on the result of prediction model generated by Tran et al. (2009), whose study found that the location of pipes determines the depth of cover and, when the cover is large, structural deterioration could be low due to less load on the pipes.

Salman and Salem (2012) demonstrated that pipe segments under local streets and alleys are less likely to deteriorate than pipe segments located under gardens or any type of roadways—a probable reason is that better design and installation practices are used for pipes installed under roads. Bakry et al. (2016b) indicated that deterioration of sewer pipes is greater when they are located under industrial zones, and lower when under residential zones. It is obvious that sewage carried by sewer pipes in industrial zones has different characteristics and causes faster deterioration.

In contrast, Tran et al. (2006) indicated that pipe location is not a significant variable in their prediction model. The effects of any critical environment, such as coastline and industrial zones, were not great enough to be considered as an influence variable. Ana et al. (2009) achieved similar results, which means that deterioration rates of sewers under light traffic and main roads were not significantly different in their model.

Soil Type

Type of soil is an important factor that can affect ground loss and stability of the sewer pipeline. For example, when soil is subjected to stress, it may behave differently due to various swell or shrinkage factors (Davies et al. 2001). Different types of soils have different reactions with pipe material, groundwater, and other pipe attributes or environmental factors (Kaushal and Guleria 2015). The fracture of soil was investigated by Davies et al. (2001), showing that sewers buried in a soil with a very high fracture potential have a significantly higher resistance to deterioration and failure. Typically, the fracture is very high in clay soils.

Wirahadikusumah et al. (2001) stated that surrounding soil is a significant factor affecting the deterioration of sewer pipes. According to this study, sewer defect size, hydraulic conditions, and soil properties are the main factors affecting the rate of ground loss. The sewer pipe can be moved when loss of the soil support occurs around the pipe. The loss of ground or soil support causes void formation around the pipe and, therefore, the sewer pipe is more likely to collapse or deform in this condition.

Soil type was also found to be a significant variable according to the result of the Markov model of Micevski et al. (2002). This study revealed that pipes in alluvial soils deteriorate faster than those in podzolic soils. Alluvial soils are deposited from a saline environment and have much more corrosive properties, while podzolic soils are formed through the weathering of rocks. The significance level of soil type might be the result of the different formation of these soil types. In contrast, soil type was insignificant in the prediction model developed by Laakso et al. (2018); in this study, the quality of soil data was not sufficient enough in their dataset, and

more research is needed to evaluate the effect of soil type on the deterioration of sewer pipes.

Groundwater Level

Groundwater is the water beneath the ground surface. The availability of groundwater at or above sewer pipelines may cause infiltration into the pipe, increasing the structural defects, formation of void, and loss of sewer support. In cohesive soil, raising the groundwater level may cause a reduction in the soil cohesive strength, and the void around the pipe can grow. Consequently, supporting soil can be washed (lost) easily and the pipe is more likely to collapse in this condition. Typically, sewers located in areas subjected to high groundwater are more likely to be at risk of failure than sewers located in areas where the groundwater level is below sewer level.

Davies et al. (2001) noted that the availability of groundwater around the pipe causes the loss of soil support and infiltration defects. Additionally, the formation of voids and lack of proper support around the pipe lead to sewer structural problems. The periodic water table in a cohesive soil may result in a reduction of soil strength and the possibility of soil being washed into the sewer.

Malek Mohammadi et al. (2019b) found that groundwater level is a significant variable based on the prediction model developed for the City of Tampa. The study revealed that groundwater increased the amount of load on the pipe and the risk of soil movement and infiltration. Typically, groundwater level is unavailable in pipeline inventories, and it is used as a determinant variable in few prediction models. More research is needed to evaluate the effect of groundwater level on condition of sanitary sewer pipes.

Summary of Results

Published studies for prediction models from 2001 through 2019 were identified and analyzed in this paper. The main objective of this research was to investigate the potential factors affecting the condition of sewer pipes, and the results are summarized in Fig. 5. It quickly became obvious that pipe age, material, diameter, depth, length, and slope were the most common variables used to develop prediction models. This result shows that municipalities and utility companies tend to collect the data that are related more to physical characteristics of sewer pipes; collecting data on physical attributes of pipes is less costly and easier, therefore they are more available in pipe inventories. Most of the sewer databases did not include any information about the soil and environmental factors around the pipes. Collecting environmental information is time consuming and expensive, and often municipalities and cities cannot allocate the budget for it. However, based on the results of reviewed studies in this paper, environmental variables can affect the condition of pipes and increase or decrease the deterioration rate of sewer pipes.

In addition to analyzing type and number of input variables, the significance level of each variable was also investigated. Most of the condition prediction studies calculated the significance level of variables. Statistical significance specifies how likely or unlikely a variable is to affect the condition of sewer pipes. The “significant” variables in Fig. 5 are shown with black circles and the “insignificant” variables are indicated with white circles. Those variables lacking any information on significance level are shown with triangles as “not specified.” In most of the studies reviewed in this paper, pipe age, material, diameter, and length are variables that significantly affect the deterioration of sewer pipes. As explained earlier, pipe age was found to be one of the most important variables affecting the condition of sewer pipes. Normally, all the

(●: significant factors, ○: insignificant factors, ▲: not specified)

Authors	Year	Independent Variables																	
		Age	Material	Diameter	Depth	Length	Slope	Sewer type	Location	Up invert	Down invert	Soil type	Bedding type	Groundwater	Corrosivity	Road type	No. Trees	Traffic	Flow
Davies et al.	2001	○	●	●	○	●		●						●	●	○	●		▲
Ariaratnam et al.	2001	●	○	●	○			●											
Wirahadikusumah et al.	2001		●		●							●		●					
Micevski et al.	2002		▲	▲								▲							
Najafi and Kulandaivel	2005	▲	▲	▲	▲	▲	▲	▲											
Jeong et al.	2005	●	○	●		●	●												
Tran et al.	2006			○	○		○		○			○					○		●
Koo and Ariaratnam	2006	●																●	▲
Tran et al.	2007	○		●	○		●		○			○					●		▲
Chughtai and Zayed	2008	●	●	○	●	●	●						●			●			
Ana et al.	2009	●	●	○	○	●	○	○	○										
Tran et al.	2009	○		●	○		○		●			○					○		●
Khan et al.	2010	●	●	●	●	●							●						
Lubini and Fuamba	2011	●	●	●		○	○												
Salman and Salem	2012	●	●	○	○	●	●	●								●			
Sousa et al.	2014	●	●	○	○	●	○												
Harvey and McBean	2014	▲	▲	▲	▲	▲	▲	▲		▲	▲					▲			▲
Bakry et al.	2016	▲	▲	▲	▲	▲	▲	▲				▲		▲				▲	▲
Gedam et al.	2016	●	○	○	●														
Bakry et al.	2016	●	●	●	●			●								○		●	
Hawari et al.	2016	▲	▲	▲	▲	▲			▲			▲	▲	▲					▲
Kabir et al.	2018	●	○	●	○	●	○			○	○								▲
Laakso et al.	2018	●	○	○	●	●	●	●	●							○	●	●	▲
Malek Mohammadi	2019	●	●	●	○	●	○					○		●	○			○	▲

Fig. 5. Variables included in sewer condition prediction models.

material has a certain lifetime and it is obvious that older pipes exhibit more defects, cracks, and hydraulic problems. Therefore, external factors such as heavy loading, earth movement, settlement, or third-party interference can damage older pipes more easily than newer pipes.

Sewer pipes constructed with different materials have different reactions to physical and environmental factors. For example, steel pipes are more likely to suffer from corrosion than plastic pipes. The relationship between the deterioration of sewer pipes and pipe material showed that this variable could significantly affect the condition of sewer pipes. Therefore, before performing pipe replacement or renewal, it is essential to investigate the behavior of pipes made of different materials in the network, and select the most appropriate material, considering the effects of other variables.

Pipe diameter was found the next significant variable based on the result of reviewed literature. Some condition prediction models identified that sewer deterioration rate decreases with increasing the diameter, but other studies found that smaller-diameter pipes have better behavior and less failure. It is not easy to prove that pipe diameter alone can increase or decrease the lifetime of pipelines. The combination of aging, material, length, and other factors should be considered to understand the effect of pipe diameter on condition of pipes. When a prediction model identifies pipe diameter as an insignificant variable, it is suggested to use different bins or ranges to categorize pipe diameter and then run the model again. This approach helps to find the relationship between dependent variable and pipe diameter.

Pipe length also played a critical role in deterioration of sewer pipes based on the reviewed literature. As explained earlier, in practically all sewer pipe inventories, length of pipe is recorded as pipe segment, which is the manhole-to-manhole length of multiple pipe sections. Recording the number of joints or length of each pipe section could help increase the performance of condition prediction models. Additionally, this information is very useful for better understanding the behavior of individual pipe sections in sewer networks.

Following the results of previous studies, pipe depth was identified as an insignificant variable in most of the prediction models. It seems that categorizing pipe depth into different intervals can be more effective to find correlations between this variable and the condition scores of sewer pipes. The effect of environmental variables has not been investigated comprehensively yet in condition prediction models. Because more online databases are available now, it is suggested to use a combination of pipe inventories and soil databases to build more proactive models practical in industry.

After development of condition prediction models, several optimization techniques can be used to determine the most appropriate size or length of pipes prior to design and installation. This analysis would be very helpful in maximizing the useful life of sewers and minimizing pipe defects and failures.

Conclusions and Recommendations for Future Research

Several techniques are employed to develop condition prediction models and evaluate the effects of different variables on the condition of sewer pipes. In this study, the influence of physical and environmental factors on sewer pipelines was reviewed, and condition prediction models developed from 2001 through 2019 were analyzed. It was observed that several factors, rather than just a single factor, affect the condition of sewer pipes. It is not logical to claim that a single factor causes pipe failure in sewer networks and, therefore, the effect of all internal and external variables should be considered to find the best solutions. The analysis

revealed that sewer condition prediction models are developed largely based on geographical characteristics, and these variables play different roles in the condition of pipelines. This is the major reason that various studies obtained contradictory results for pipe length, depth, or diameter factors. For example, based on the location of the city, type of soil, and/or other influence variables, smaller pipes would have better lifecycle condition than larger pipes. Thus, the factors that influence the condition of sewer pipes are numerous and widely varied in nature.

In addition, the results indicated that studying the effect of many variables on sewer pipe conditions is very difficult with data that are routinely collected by municipalities and utility companies. More intelligent and smart data collection systems are required to prepare appropriate inventories of sewer pipes. For example, several devices can be used to collect real-time data and store them in a cloud-based platform. It was observed that most of the research studies in this area have been conducted on few pipe segments, and PACP and WRC are the most common condition assessment standards based on the results of reviewed studies. The findings of this study can guide practitioners in creating customized condition assessment models for their agencies to reduce incidents and save maintenance costs.

To better understand the effect of pipe length on deterioration of sewer pipes, it is recommended that researchers focus on number of joints or length of pipe sections as an independent variable. Additionally, further investigation is required to evaluate the performance of new machine learning and deep learning algorithms for development of condition prediction models. During the past few years, several statistical and AI models were developed to predict the condition of sewer pipes; however, there is still high demand for more advanced models with more critical input variables.

Data Availability Statement

No data, models, or code were generated or used during the study.

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